Time Series Homework

Environmental Engineering

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1 River Nile at Ashwan

The `Nile` dataset contains measurements of the annual flow of the River Nile at Ashwan from 1871 to 1970. This analysis focuses on the sub-period **1889–1959**.

```r
# Subset to the period of interest, preserving the time-series structure
mynile <- window(Nile, start = 1889, end = 1959)

# Convert to a plain data frame for easier manipulation
mynile_df <- data.frame(
  year = as.numeric(time(mynile)),
  flow = as.numeric(mynile)
)
```

2 Descriptive Statistics

```r
# Flow recorded in 1919
flow_1919 <- mynile_df$flow[mynile_df$year == 1919]
cat("Flow in 1919:", flow_1919, "\n")
```

```
## Flow in 1919: 764
```

```r
# Median annual flow over the full sub-period
median_flow <- median(mynile_df$flow)
cat("Median annual flow:", round(median_flow, 3), "\n")
```

```
## Median annual flow: 848
```

```r
# Full summary for reference
summary(mynile_df$flow)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     456     778     848     888     985    1260
```

```r
# Base plot: annual flow as a blue line
plot(mynile_df$year, mynile_df$flow,
  type = "l",
  col  = "blue",
  lwd  = 1.5,
  xlab = "Year",
  ylab = expression("Discharge (m"^3*"/s)"),
  main = "River Nile at Ashwan - Annual Discharge"
)

# Horizontal line at the median level
abline(h = median_flow, col = "orange", lwd = 2)

# Label the median line
text(1889, median_flow + 25,
  paste0("Median = ", round(median_flow, 0), " m\u00b3/s"),
  col = "orange", adj = c(0, 0), cex = 0.8
)

# Mark the 1919 flow with a star
points(1919, flow_1919,
  pch = "*", col = "red", cex = 3
)
```

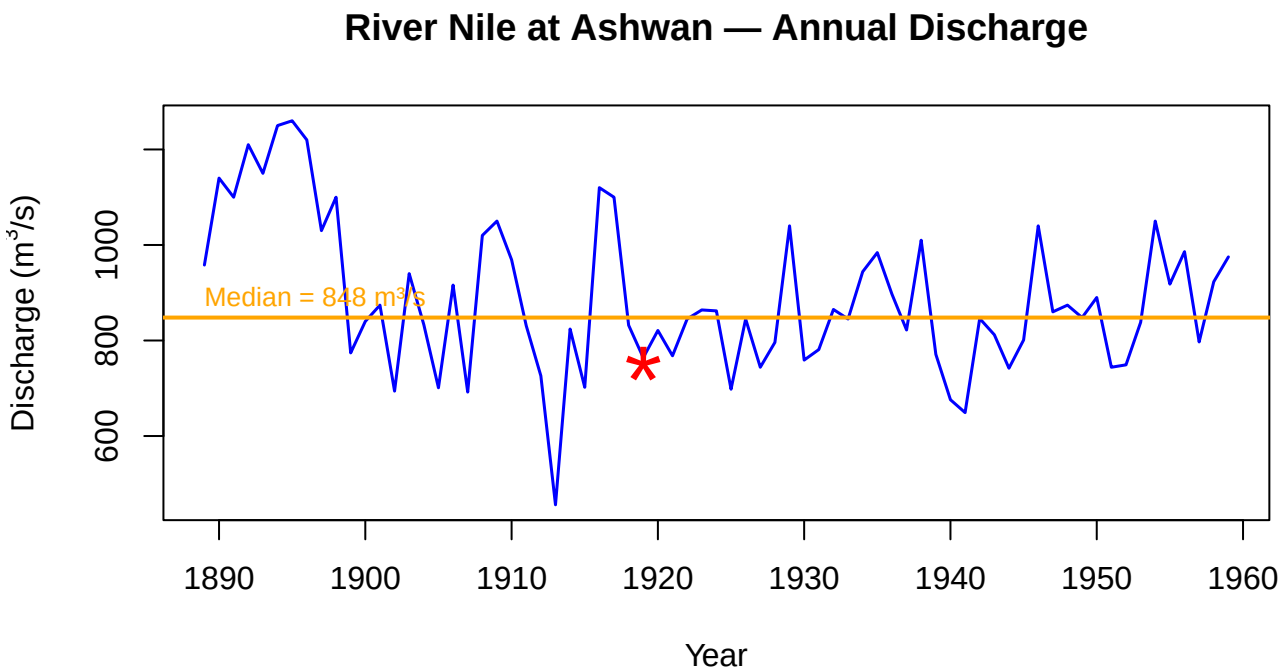

Figure 1: Annual discharge of the River Nile at Ashwan (1889–1959). Orange horizontal line: median flow. Red star: flow in 1919.

3 Linear Trend and Filtering

3.1 3-Year Moving Average

```
# Apply a symmetric 3-year moving average filter
mynile_ma3 <- filter(mynile_df$flow, filter = rep(1/3, 3), sides = 2)

# Smallest value produced by the filter (ignoring boundary NAs)
min_ma3 <- round(min(mynile_ma3, na.rm = TRUE), 3)
cat("Smallest 3-yr moving average value:", min_ma3, "\n")
```

```
## Smallest 3-yr moving average value: 661
```

3.2 Linear Trend Model

```
# Fit a simple linear regression: flow ~ year
lm_nile <- lm(flow ~ year, data = mynile_df)

# Display full model summary
summary(lm_nile)
```

```
##
## Call:
## lm(formula = flow ~ year, data = mynile_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -459.0  -89.4  -16.1  120.5  300.9
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5596.09    1673.16    3.34   0.0013 **
## year           -2.45       0.87   -2.81   0.0064 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 150 on 69 degrees of freedom
## Multiple R-squared:  0.103,  Adjusted R-squared:  0.0899
## F-statistic: 7.92 on 1 and 69 DF,  p-value: 0.00637
```

```r
# Report the estimated slope (rounded to 3 d.p.)
slope <- round(coef(lm_nile)["year"], 3)
cat("Estimated slope (year coefficient):", slope, "\n")
```

```
## Estimated slope (year coefficient): -2.45
```

```r
# Reproduce the base plot
plot(mynile_df$year, mynile_df$flow,
  type = "l",
  col  = "blue",
  lwd  = 1.5,
  xlab = "Year",
  ylab = expression("Discharge (m"^3*"/s)"),
  main = "River Nile at Ashwan - Trend and Filtering"
)

# Median reference line
abline(h = median_flow, col = "orange", lwd = 2)

# 3-year moving average (dotted red)
lines(mynile_df$year, mynile_ma3, col = "red", lty = 2, lwd = 2)

# Linear trend (solid green)
abline(lm_nile, col = "green", lwd = 2)

# 1919 star marker
points(1919, flow_1919, pch = "*", col = "red", cex = 3)

# Legend
legend("topright",
  legend = c("Annual flow", "Median", "3-yr moving avg", "Linear trend"),
  col    = c("blue",        "orange", "red",              "green"),
  lty    = c(1,              1,        2,                   1),
  lwd    = c(1.5,            2,        2,                   2),
```

```
  cex     = 0.8
)
```

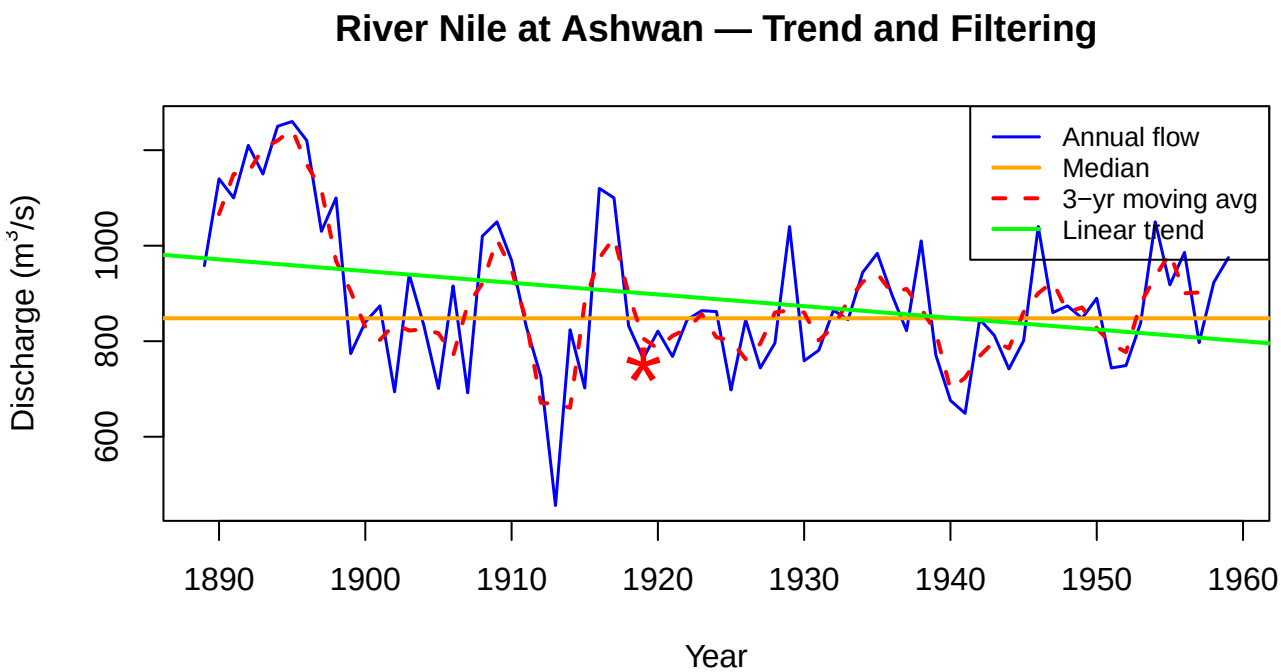

Figure 2: Annual discharge with 3-year moving average (dotted red) and linear trend (green). Orange line: median.

---

4 Autocorrelation

4.1 ACF — Autocorrelation Function

```
# Compute and plot the ACF
acf_result <- acf(mynile_df$flow, plot = TRUE,
  main = "ACF - River Nile at Ashwan")
```

```
# Autocorrelation at lag 3 (index 4 because lag-0 is stored at index 1)
lag3_acf <- round(acf_result$acf[4], 3)
cat("Autocorrelation at lag 3:", lag3_acf, "\n")
```

```
## Autocorrelation at lag 3: 0.218
```

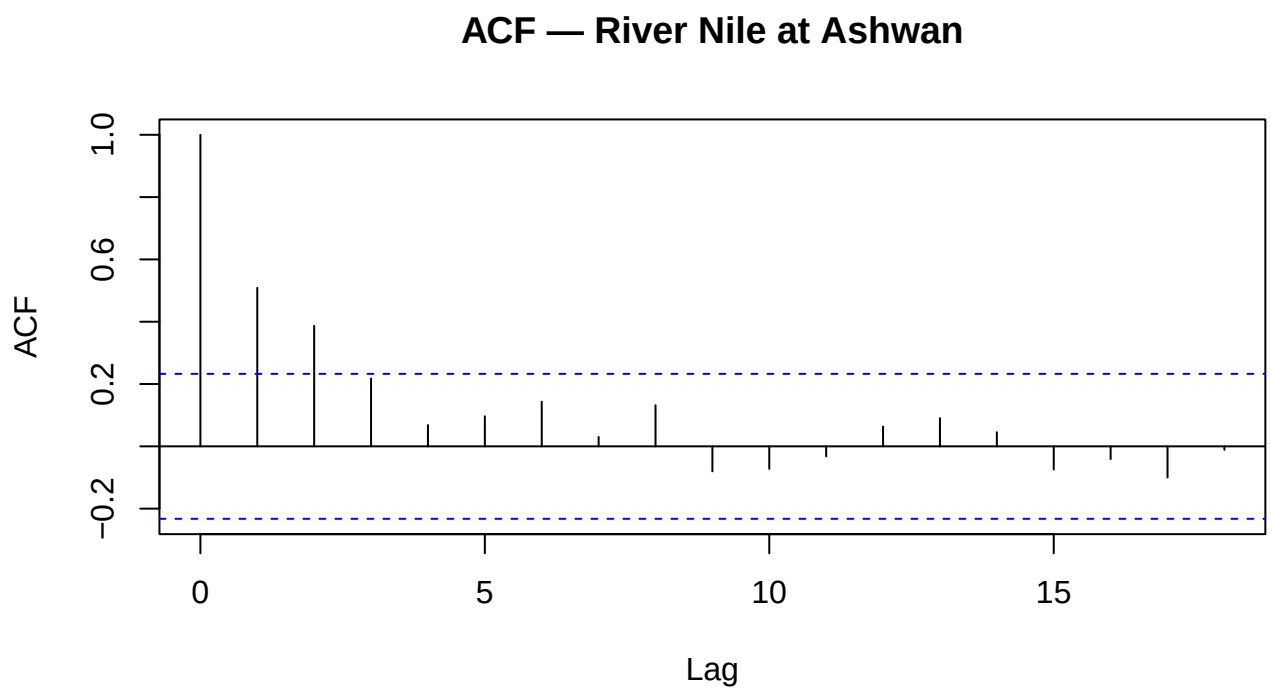

Figure 3: Autocorrelation function (ACF) of the Nile sub-series.

```
# Smallest lag (>0) with negative autocorrelation
# acf$acf[1] is lag 0, so lag k is at index k+1
neg_lags <- which(acf_result$acf < 0) - 1   # convert index to lag
min_neg_lag <- min(neg_lags[neg_lags > 0])
cat("Smallest lag with negative autocorrelation:", min_neg_lag, "\n")
```

```
## Smallest lag with negative autocorrelation: 9
```

4.2 PACF — Partial Autocorrelation Function

```
# Compute and plot the PACF
pacf_result <- pacf(mynile_df$flow, plot = TRUE,
  main = "PACF - River Nile at Ashwan")
```

```
# Partial autocorrelation at lag 6
lag6_pacf <- round(pacf_result$acf[6], 3)
cat("Partial autocorrelation at lag 6:", lag6_pacf, "\n")
```

```
## Partial autocorrelation at lag 6: 0.137
```

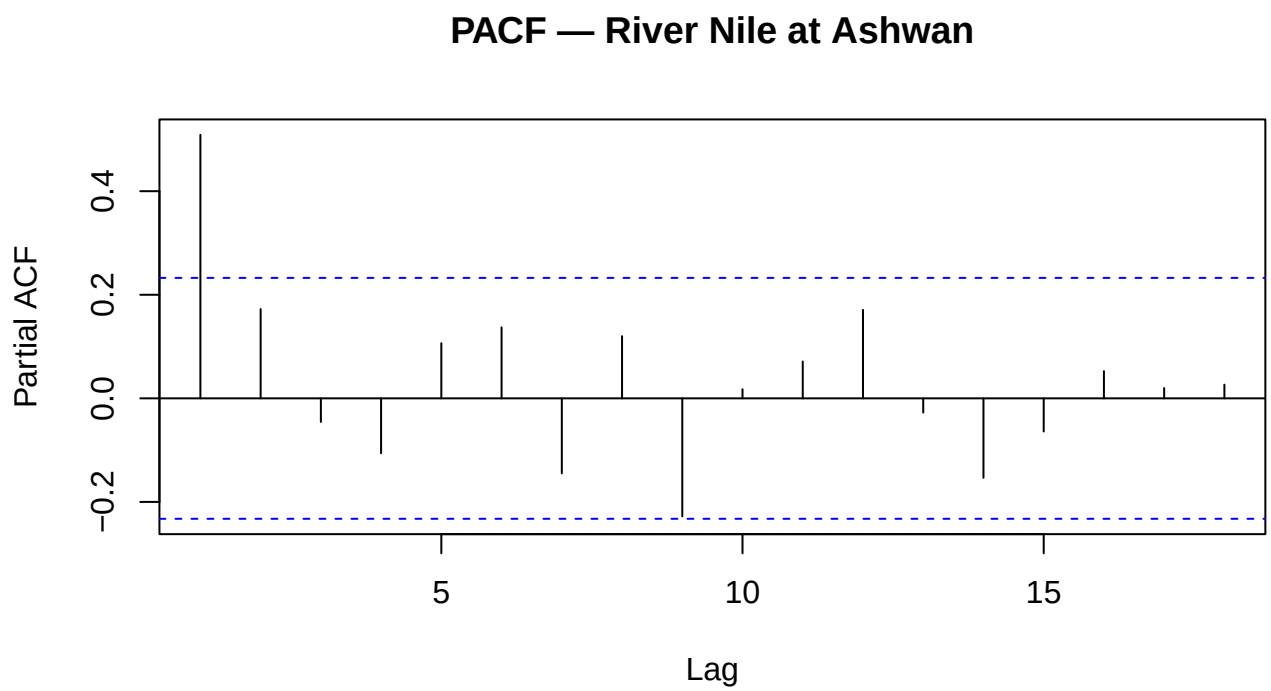

Figure 4: Partial autocorrelation function (PACF) of the Nile sub-series.

5 ARIMA Models

5.1 Model Fitting — ARIMA(3,2,2)

```
# Fit an ARIMA(3,2,2) model: AR order 3, differencing order 2, MA order 2
arima322 <- arima(mynile_df$flow, order = c(3, 2, 2))

# AR1 coefficient (Alpha1)
ar1_coef <- round(arima322$coef["ar1"], 3)
cat("AR1 coefficient (alpha_1):", ar1_coef, "\n")
```

```
## AR1 coefficient (alpha_1): 0.311
```

```
# AIC of the fitted model
aic_val <- round(AIC(arima322), 3)
cat("AIC:", aic_val, "\n")
```

```
## AIC: 892
```

5.2 Model Diagnostics

```r
# Standard diagnostic plot: residuals, ACF, and Ljung-Box test p-values
tsdiag(arima322)
```

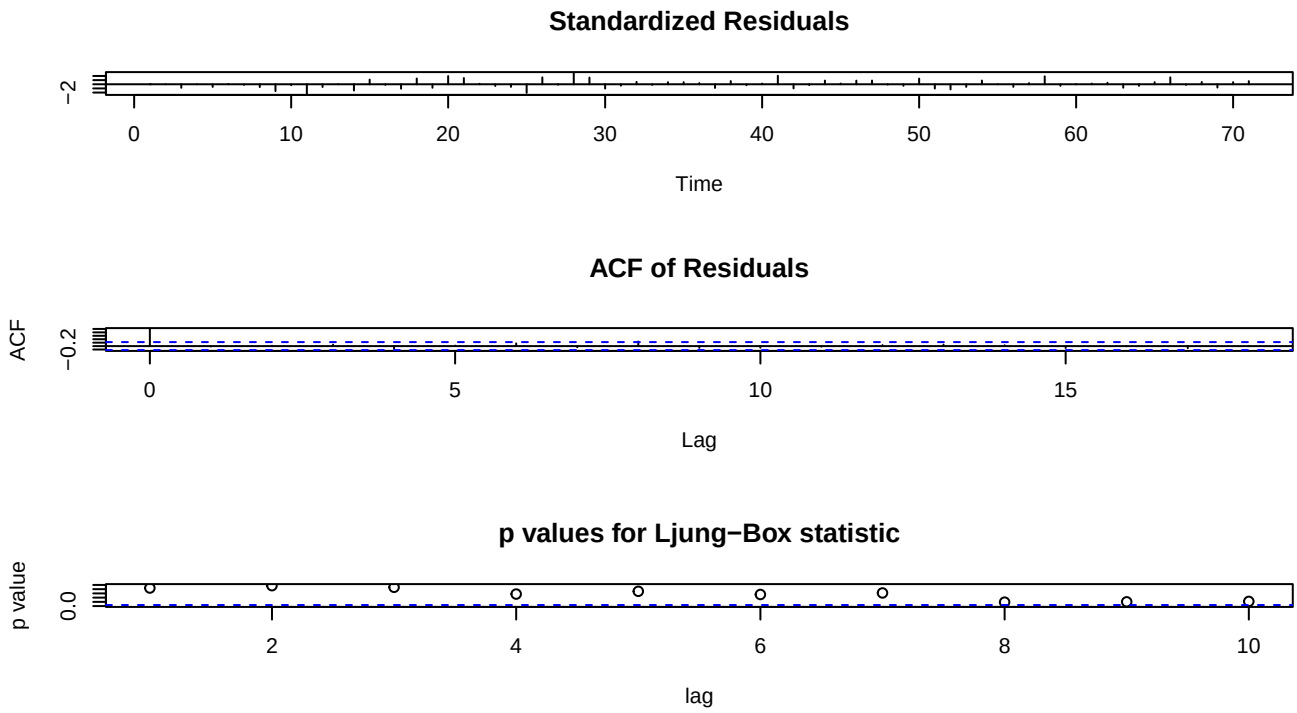

Figure 5: ARIMA(3,2,2) model diagnostics: standardised residuals, ACF of residuals, and Ljung–Box p-values.

5.3 Forecasting

```r
library(forecast)

# Generate a 3-year ahead forecast and plot with prediction intervals
plot(forecast(arima322, h = 3),
  main = "ARIMA(3,2,2) Forecast - River Nile at Ashwan",
  xlab = "Year",
  ylab = expression("Discharge (m"^3*"/s)")
)
```

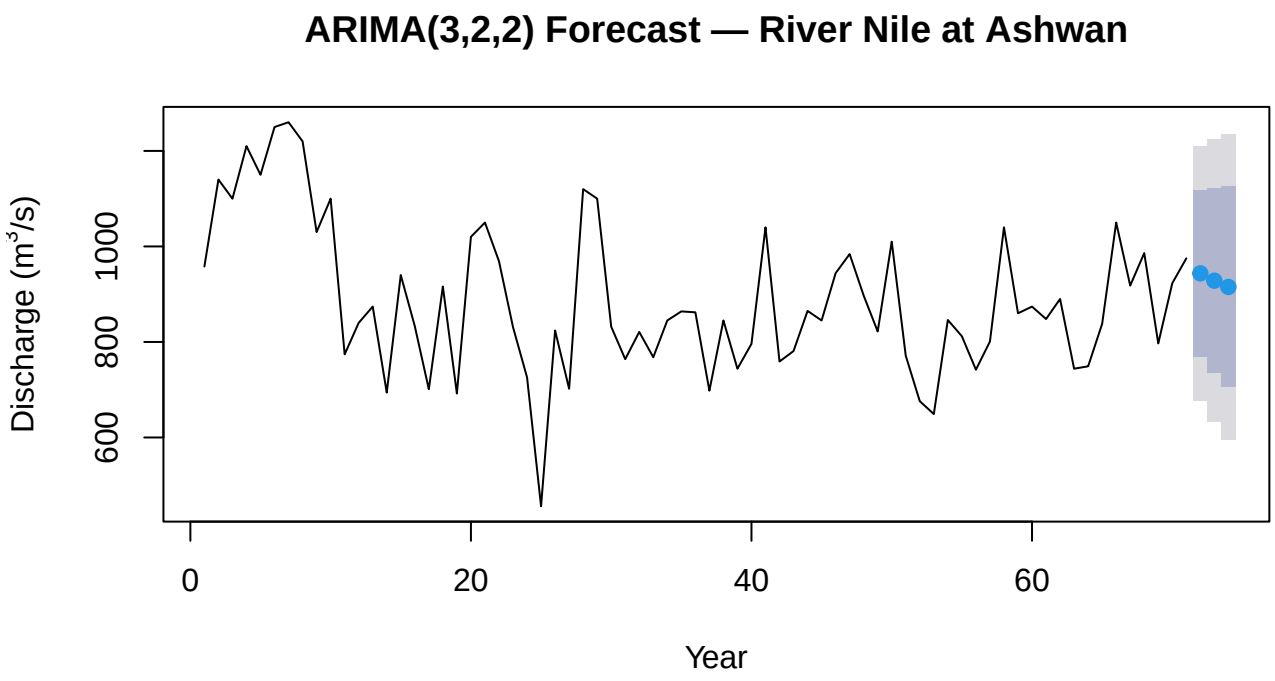

Figure 6: 3-year ahead forecast from the ARIMA(3,2,2) model with 80% and 95% prediction intervals.